

AI: RESPONSIBILITY IN A CHANGING WORLD

Prof. Dr. Virginia Dignum

Chair Responsible AI – Director AI Policy Lab, Umeå University

Guest professor SIAS, University of Amsterdam

Email: virginia@cs.umu.se



UMEÅ UNIVERSITY



AI IN 2026 – ISSUES WE FACE

CONCIOUSNESS?



AI FIRST?

WHO ARE WE?



CEOs



WHAT DO WE WANT?



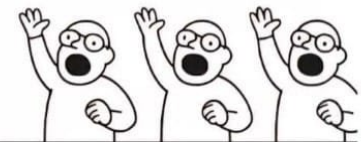
AI!



AI TO DO WHAT?



WE DON'T KNOW!



WHEN DO WE WANT IT?



RIGHT NOW!



THE AI FIRST APPROACH

- Moves **AWAY FROM** governance TO **unchecked deployment**
- Ignores evidence of limited productivity gains from AI
- **Missing "Why AI?"** and **"Whether AI?"** questions
- Democratic deficit: lacks stakeholder consultation
- Creates **pressure to adopt without evaluation** (Dignum et al., 2025)

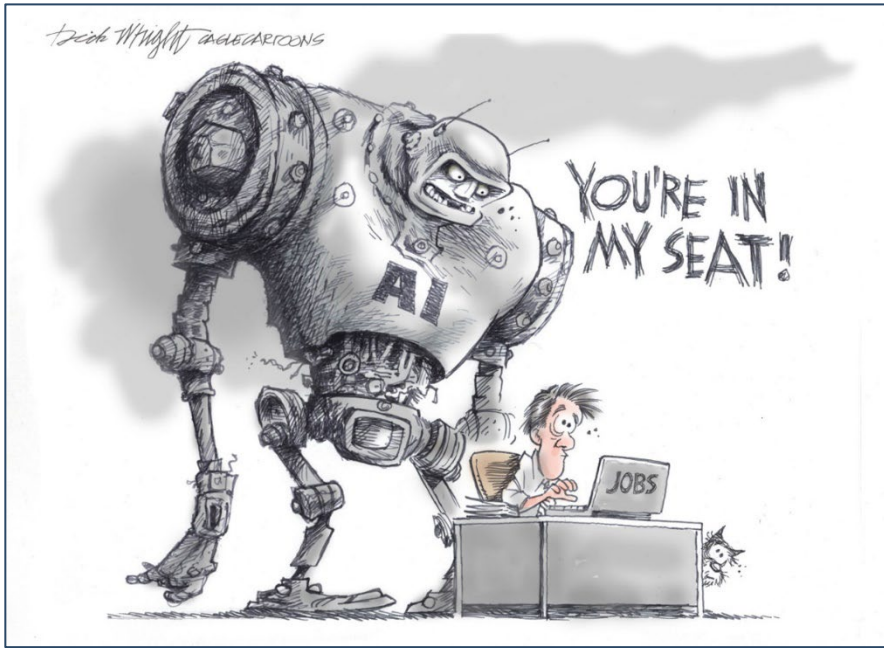
Incentive structures that promote AI use regardless of suitability will backfire: municipalities and regions need competence to assess when and where AI is appropriate, not structures that drive adoption indiscriminately

Rests on misconception that regulation impedes digital services

AI adoption without domestic R&D investment deepens dependency on a small number of foreign providers, undermining the very digital sovereignty the strategy claims to advance.

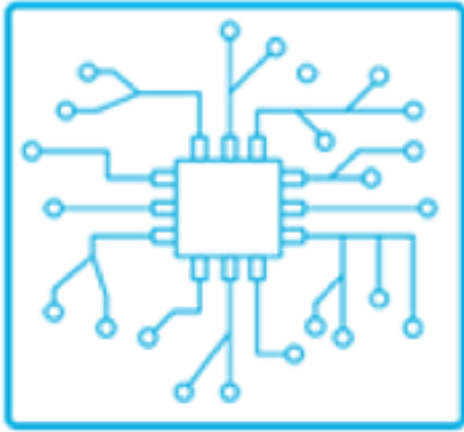
Accountability follows control. Producers design the system; they should bear primary responsibility. Offloading verification to users merely relocates liability without reducing risk.

ARE WE SHAPING AI, OR IS AI SHAPING US?



- AI raises as many concerns as it solves
- AI is not a means to replace humans
- Need human insight to address AI's ethical, social, and unintended consequences

AI – MORE IS BETTER?



Computing Power

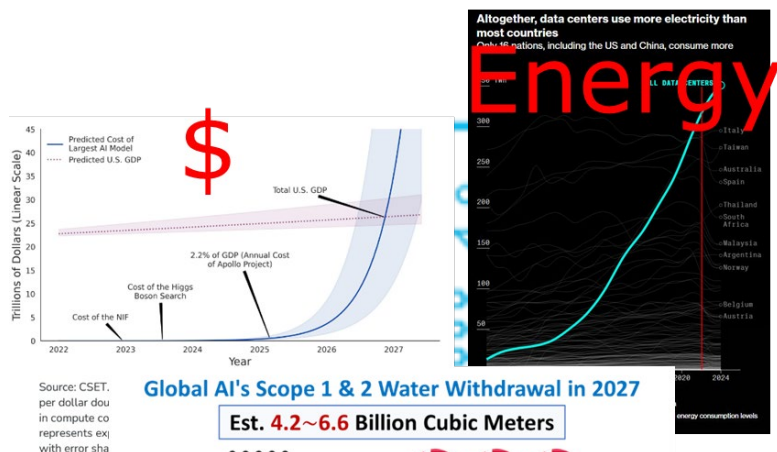


Algorithm Power



Data Availability

AI – MORE IS BETTER?

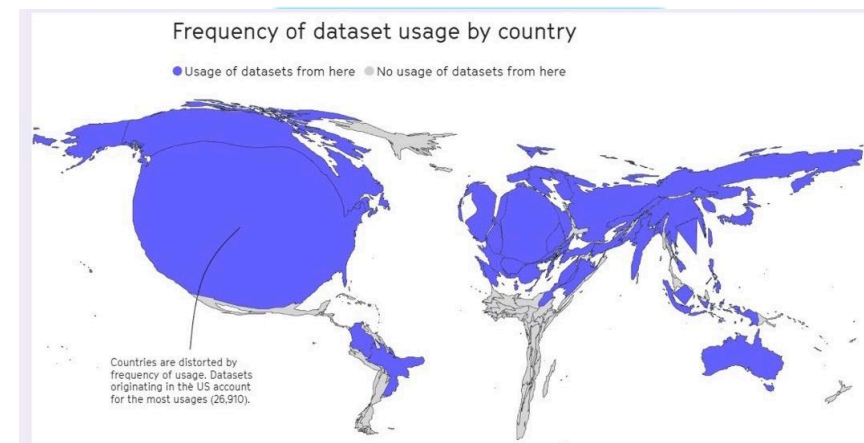


AI ≈ 4~6x Annual Water Withdrawal of Denmark

Computing Power



Algorithm Power



Data Availability

Overcoming Racial Bias In AI Systems And Startlingly Even In AI Self-Driving Cars

AI expert calls for end to UK use of 'racially biased' algorithms

Gender bias in AI: building fairer algorithms

Bias in AI: A problem recognised but still unresolved

Millions of black people affected by racial bias in health-care algorithms

Google exploited homeless black people to develop the Pixel 4's facial recognition AI

Russia Tests New Disinformation Tactics in Africa to Expand Influence

Amazon's facial recognition matched 28 members of Congress to criminal mugshots

Flawed Algorithms



Racial patient

AI E Live Reg

Amazon, Apple transcribing bl AI voice recogni

When It Comes to Gori



WILL A ROBOT STEAL YOUR JOB?

KILLER COMPUTERS

Bill Gates warns 'dangerous AI' poses a threat 'like nuclear weapons'

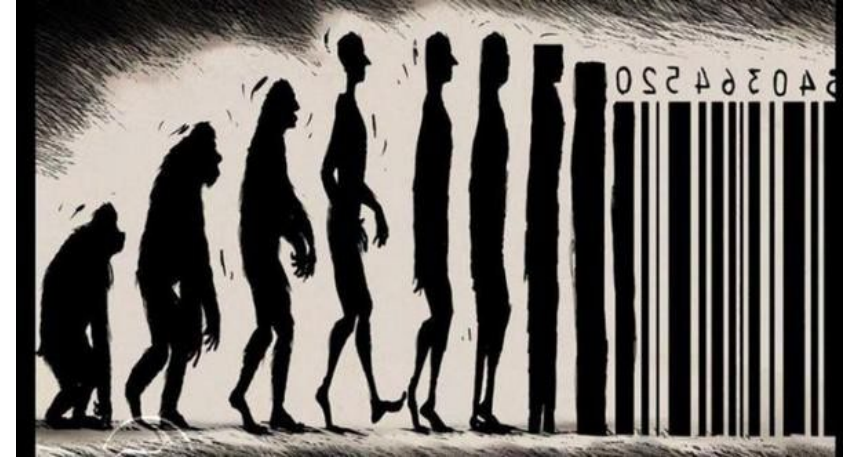
AI WARNING:
Robots will destroy a HUGE number of jobs, claims expert

AI could be used to **TAKE OVER** the **WORLD** through 'evil' fake news and hijacking cars

AI IMPACT MEANS HUMAN RESPONSIBILITY

RESPONSIBLE AI – WHY CARE?

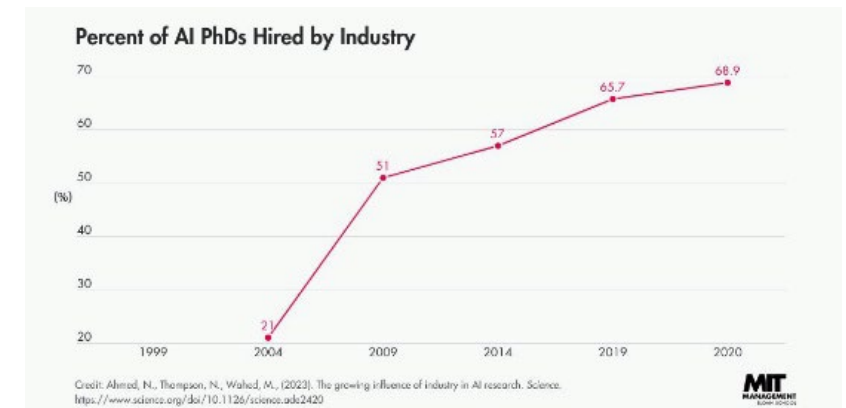
- **Datification**
 - Reality is more than data
 - Data is constructed
 - Data is biased
 - Data rich, insight poor
- Power
- Sustainability



RESPONSIBLE AI – WHY CARE?

- Datification
- Power
 - Who is developing AI?
 - Who is deciding?
- Sustainability

At Tech's Leading Edge, Worry About a Concentration of Power



- **18% researchers at conferences are women**
- **80% professors are men**
- **Workforce**
 - **Google: 2,5% black, 3,6% Latino, 10% women**
 - **Facebook: 3,8% black, 5% Latino, 15% women**

RESPONSIBLE AI – WHY CARE?

- Datification

- Power

- Sustainability

- Computational cost of AI
- Human and social costs

INDEPENDENT PREMIUM

THE LONGER READ

Behind the AI boom, the armies of overseas workers in ‘digital sweatshops’



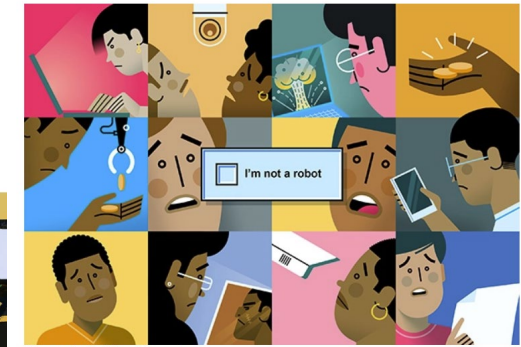
Technology

Turk Wars: How AI Threatens the Workers Who Fuel It

The much-hyped AI tools of the future are being built by a globally dispersed army of data workers.

CITE SHARE COMMENT PRINT ORDER REPRINTS

By Krystal Kauffman & Adrienne Williams | Oct. 11, 2023



OpenAI Used Kenyan Workers on Less Than \$2 Per Hour: Exclusive | TIME

AI IS NOT NEUTRAL

- AI can advance human dignity, or undermine it
- embodies human choices
- Governing AI = Governing ourselves



AI – AI ETHICS – RESPONSIBLE AI – AI GOVERNANCE...

- Early focus
 - Technical performance and efficiency of AI systems
- Growing awareness
 - Recognition of risks such as bias, discrimination, privacy breaches, lack of accountability
- Shift to ethics
 - Calls for fairness, transparency, human oversight, and explainability
- Responsible AI
 - Moving from principles to practice by embedding ethics in design, deployment, and use
- Governance need
 - Ensuring societal trust, long-term sustainability, and alignment with democratic values
- Global momentum:
 - AI ethics guidelines, regulatory proposals (e.g., EU AI Act), and cross-sector initiatives
 - One step forward, two steps backwards

WHAT WE ARE TALKING ABOUT?

- **Bias:** systematic deviation, not a synonym for error.
 - Important distinction: statistical bias (sampling, measurement) is not the same as societal bias (the system reproduces an unjust pattern).
- **Fairness:** contested distributive claim, not a metric.
 - Formal criteria (parity, equalised odds, calibration) are mutually incompatible except in degenerate cases, so choosing one is a normative act.
(Kleinberg et al. 2016; Chouldechova 2017)
- **Transparency** (about process and provenance) is different from explainability (about a specific output).
 - They are different obligations with different audiences.
- **Privacy** is about contextual integrity, not only data protection compliance
(Nissenbaum)
- **Accountability:** answerability plus attributability:
 - There is someone that can be asked *and* must be able to answer.
- **Robustness** and **validity:** does the model measure what it claims, under distribution shift.

WHAT DOES IT MATTER FOR CLIMATE?

- **Problem choice and scoping**
 - What counts as the problem, and whose problem it is, is decided before any data is collected.
 - E.g. "Predict yield" and "support a smallholder's decision" are not the same problem.
- **Target variable and proxies**
 - The label is rarely the thing you care about. damage claims as a proxy for flood impact, grid-connected demand as a proxy for energy need.
- **Data coverage**
 - Models are validated where data already exists, and inferred where it does not.
 - In much of Africa weather station coverage is sparse, and stations cluster in cities and along major roads, so coverage is worst in rural areas where livelihoods are most climate-exposed.
- **Label provenance and transferability**
 - A model can be technically excellent and still be wrong everywhere it was not trained.
- **Evaluation choices**
 - Aggregate skill scores hide regional and seasonal failure. Disaggregate, or you will not see it.
- **Deployment and feedback**
 - Risk maps reallocate insurance, credit and investment, changing the exposure they claim to measure.
 - Observation-based flood exposure is roughly ten times higher than earlier model-based estimates
 - unmapped areas are not unaffected areas.

Fix is not a better model. It is better questions, earlier.

RAI FOR CLIMATE AI – EXERCISE 1

Scenario: A city commissions a flood risk model. It is trained on twenty years of insurance damage claims: each recorded claim marks a location as having been flooded, and locations with no claim are treated as not flooded. The model uses satellite imagery, elevation and rainfall as inputs, and reaches 91% accuracy on held-out data.

Where does bias enter first?

- The satellite imagery resolution
- The choice of insurance claims as the label
- The model architecture
- The evaluation metric



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RAI FOR CLIMATE AI – EXERCISE 1

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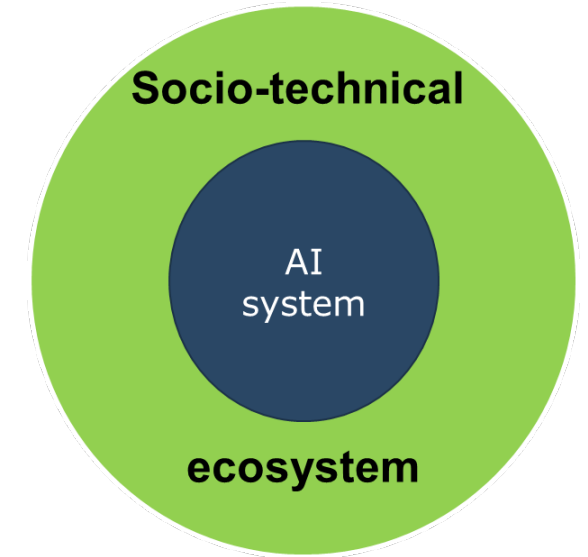
- The satellite imagery resolution
- The choice of insurance claims as the label
- The model architecture
- The evaluation metric

Claims record who was insured and who filed, not who was flooded. Uninsured households and informal settlements are not underrepresented, they are recorded as dry. The 91% is measured against the same flawed label, so it cannot detect the problem. Everything downstream inherits it.

RESPONSIBLE ABOUT AI

AI does not happen to us!

- AI is designed. We make the choices
 - Who should decide?
 - Which values should be considered?
 - Whose values? How to prioritize?
- Ask Question Zero!
 - Is AI the best option here?
 - Ask why before asking how
- AI does not exist in a vacuum
 - There is no technology fix for ill effects!
 - Ethics, regulation, governance concern the ecosystem



Responsible AI solutions need to be social rather than technical!

THE ALTERNATIVE – PURPOSE FIRST (QUESTION ZERO)

Purpose First approach asks: "**Whether and where** is AI actually needed and beneficial?"

"Should this AI system be built at all?"

Key Principles:

- Start with **societal needs**, not technology capabilities
- Ask systematically: **Why? For whom? What are alternatives?**
- Include **diverse voices** in **decision-making**
- Focus on direction and public value, not speed

Question Zero:

- Shifts focus from SPEED to **DIRECTION**
- Empowers diverse actors (SMEs, civil society, small countries)
- Provides practical, low-cost framework
- Grounds decisions in real **problem identification** (Dignum et al., 2025)

Download: https://aipolicylab.se/wp-content/uploads/2026/04/Q0-Assessment-Tool-v3_draft_march-2026.pdf

QUESTION ZERO GOES FURTHER THAN GO/NO-GO

The question Zero lifecycle:

- **Framing**: whose problem, who defined it, what is the non-AI baseline, who bears the cost of a wrong answer.
- **Data**: provenance, consent, coverage gaps, who is invisible in the data, under what terms was it obtained.
- **Modelling**: what is optimised, what is traded away, what uncertainty is communicated.
- **Evaluation**: disaggregated performance, not aggregate; failure modes named, not just accuracy.
- **Deployment**: who operates it, what recourse exists, what happens when it is wrong.
- **Monitoring and exit**: who maintains this after the project/grant ends, and how is it decommissioned.

EXERCISE 2: AI FOR FLOOD ADAPTATION PLANNING

Scenario: A city wants to introduce an AI system that ranks neighbourhoods for flood adaptation funding

- The system predicts flood risk from satellite imagery, elevation data and past damage claims, and ranks areas for investment.
- The system estimates household vulnerability by combining remote sensing with mobile phone and utility data.
- The system generates summaries and recommendations for the council, which has three weeks to decide.

QUESTION ZERO – EXERCISE 2

- Is AI suitable for this purpose?
- Options
 - A) Yes, the alternative is a slower and equally biased manual process.
 - B) Yes, the alternative is a slower and equally biased manual process.
 - C) No, the data cannot support the decision it is being used for.
 - D) No, the data cannot support the decision it is being used for.



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DESIGN IS POLITICAL

- Choices
- Formulation
- Information
- Involvement
- Legitimacy
- Aggregation



WHAT DO WE WANT AI TO BE?

WHAT IS AI?

- Human-like?
 - Why?
 - What does this mean?
- Tool?
 - For what? For whom?
- Simulation or operation?
 - Understand intelligence by building intelligence, or
 - Active intervention in real world
- Normative or descriptive?
 - Do as we say or do as we do?



PRINCIPLES, GUIDELINES, ETHICS

GOVERNANCE – WHY? WHAT FOR?

- Regulation as **incentive for responsible innovation, sustainability, and fundamental human rights**
 - powerful stepping stone for innovation with societal benefits
 - signaling expected ambitions enhancing innovation, competitive power

Cars drive faster with brakes

-

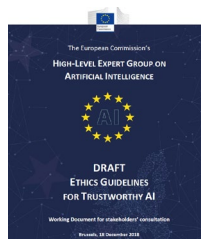
In a game without rules, no one wins

- Ensuring safety, security, standards, and governance approaches to AI systems, products, and results
- Need for better understanding and integration of existing frameworks alongside introducing more regulation
- Avoidance of an "arms race" narrative in AI regulation

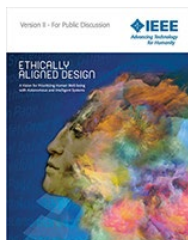
PRINCIPLES AND GUIDELINES

EU HLEG	OECD	IEEE EAD
- Human agency and oversight Technical robustness and safety - Privacy and data governance Transparency Diversity, non-discrimination and fairness Societal and environmental well-being Accountability	benefit people and the planet - respects the rule of law, human rights, democratic values and diversity, - include appropriate safeguards (e.g. human intervention) to ensure a fair and just society. transparency and responsible disclosure robust, secure and safe - Hold organisations and individuals accountable for proper functioning of AI	- How can we ensure that A/IS do not infringe human rights? - effect of A/IS technologies on human well-being. - How can we assure that designers, manufacturers, owners and operators of A/IS are responsible and accountable? - How can we ensure that A/IS are transparent? - How can we extend the benefits and minimize the risks of AI/AS technology being misused?

Level	Framework & Reach
Global	UNESCO (194 countries), OECD (>70 jurisdictions), GPAI (25+ members), CoE treaty (50+ countries), G7 principles
Regional	EU AI Act (27 EU states), Santiago Declaration (Latin America/Caribbean)
National	930+ initiatives in 71 countries
Others	200+ guidelines across NGOs, academic, private bodies



<https://ec.europa.eu/digital-single-market/en/high-level-expert-group-artificial-intelligence>



<https://ethicsinaction.ieee.org>

OECD Principles on Artificial Intelligence



<https://www.oecd.org/digital/ai/principles/>



<https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>



<https://www.un.org/ai-advisory-body>

Well over 1000 published worldwide!

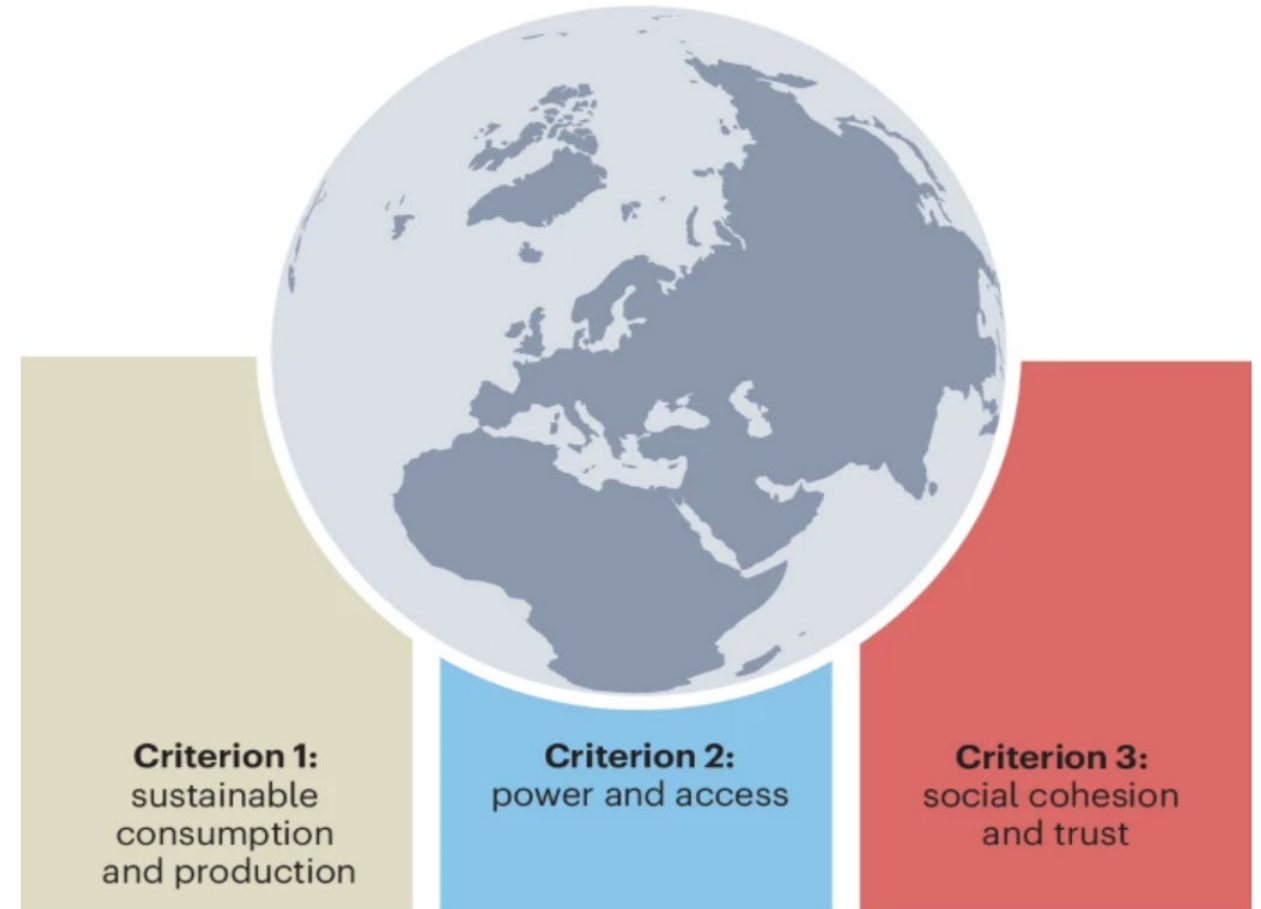
SAFEGUARDING DEMOCRACY

- Strengthening Regulation and Oversight
 - Implement clear laws for AI, facial recognition, cryptocurrency, and spyware.
 - Enhance transparency and accountability in technology deployment.
 - Strengthen democratic institutions to oversee and guide technological development.
- Promoting Public Participation
 - Democratize AI development by involving diverse societal stakeholders.
 - Align technology systems with democratic values and public interest.
- Enhancing Digital Literacy
 - Foster critical understanding of AI and digital technologies through education.
 - Empower citizens to advocate for their rights in the digital age

SAFEGUARDING THE PLANET

AI impact demands systemic stewardship, beyond environment impacts:

1. help to accelerate the **transition to sustainable** production and consumption in ways that respect planetary boundaries, or at least do not obstruct these objectives
2. be developed, deployed and used in ways that **ensure equitable access** to AI tools for global sustainability and avoid concentrations of power
3. be developed, deployed and used to **support greater societal cohesion**, build trust and provide access to reliable information for planetary stewardship



Owen Gaffney, Amy Luers, Franklin Carrero-Martinez, Berna Oztekin-Gunaydin, Felix Creutzig, Virginia Dignum, Victor Galaz, Naoko Ishii, Francesca Larosa, Maria Leptin & Ken Takahashi Guevara. The Earth alignment principle for artificial intelligence. *Nat Sustain* (2025). <https://doi.org/10.1038/s41893-025-01536-6>

MEASURING IMPACT: WHAT COUNTS, AND WHO ANSWERS FOR IT

We govern what we count. Most of AI's impact is not counted.

Three levels of impact

(Kaack et al., 2022)

- Compute
- Application
- System

The governance consequence

- Accountability follows measurement
 - compute has an owner, system effects often have none
- What is reported becomes what is designed for
 - Choosing the metric is a policy act, not a technical one
- Earth alignment
 - planetary boundaries, equitable access, societal cohesion.
- Question Zero applies to metrics too
 - agree what counts, and who audits it, before building

Kaack et al., Nature Climate Change 12, 518-527 (2022) ·
Gaffney et al., Nature Sustainability 8, 467-469 (2025)

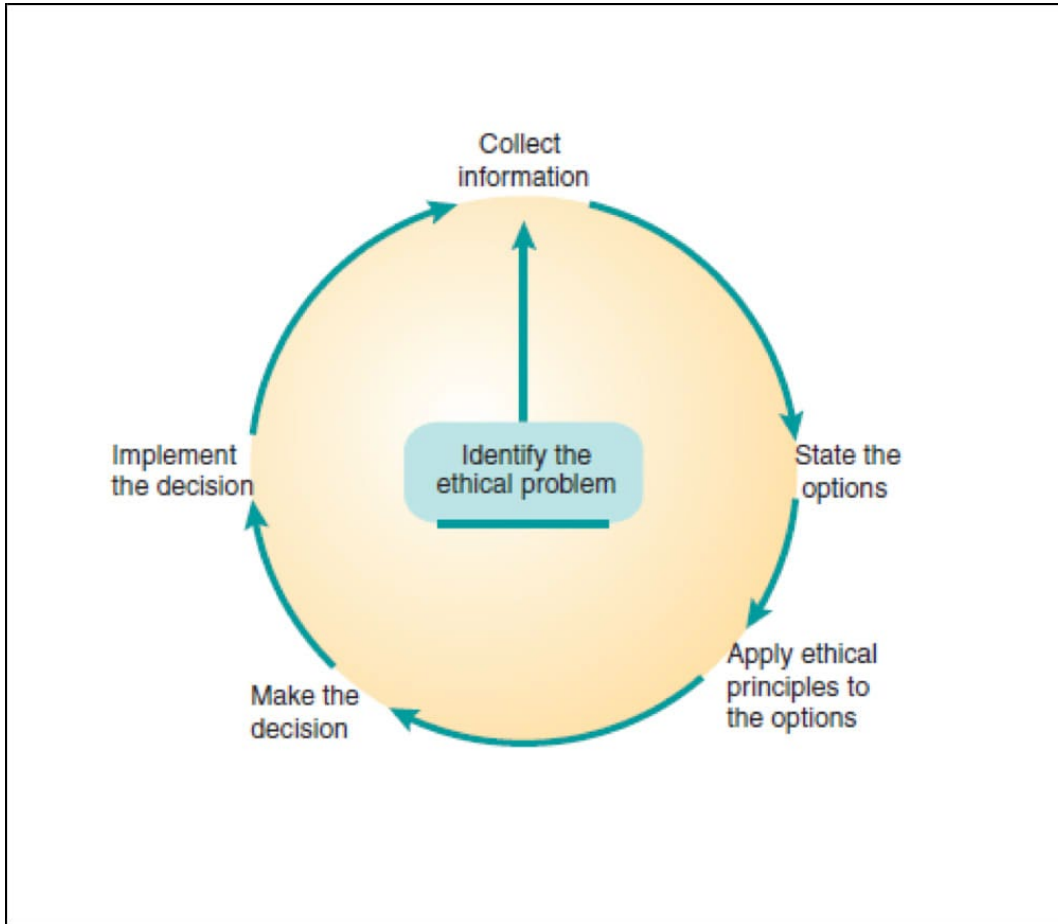
CLIMATE JUSTICE

- distributive (who gets the benefit and who carries the compute, water, land and labour cost),
 - procedural (who was in the room),
 - recognitional (whose knowledge counts as evidence).
-
- Ends justify means? Tradeoffs?
 - E.g. is improving fossil fuel extraction a responsible use of AI?
 - What are the alternatives? Who benefits? Who loses?

OPERATIONALISATION: REGULATION AND MORE

- Regulation
 - AI Act: Human-centered, risk-based approach
- Standards
 - soft governance; non mandatory to follow
 - demonstrate due diligence and limit liability
 - user-friendly integration between products
- Organisation structures and procedures
 - Advisory boards and ethics officers
 - Set and monitor ethical guidelines
- Assessment for trustworthy AI
 - responsible AI is more than ticking boxes
 - Means to assess maturity are needed
- Awareness and Participation
 - Education and training
 - Appeal to civic duty / voluntary implementation

ETHICAL REASONING



- Not about the answer but about recognizing the issue
- Not one answer! No silver bullet!
 - Utilitarian view
 - Rights view
 - Justice view
 - Common good view
 - Virtues view
 - ...

ETHICS IN REAL LIFE

- Fairness
 - Child support subsidy - equity
 - Parental leave - equality
- Safety
 - Speed sensors – short term
 - Traffic lessons for children – long term
- Fairness or safety
 - Limited resources

DILEMMAS – EXERCISE 3

- **Climate adaptation grants:** you were asked to develop an AI system to screen applications to a municipal climate adaptation fund, awarding grants based on each application's likelihood to deliver "high climate impact per euro" using data from previously funded projects. This replaces the need for manual review and site visits by case officers.
- Trade-offs: how to split your resources:
 - How much (computational) cost is worth 5% more accuracy?
 - How much (computational) cost is worth to include explanation?



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AI ETHICS

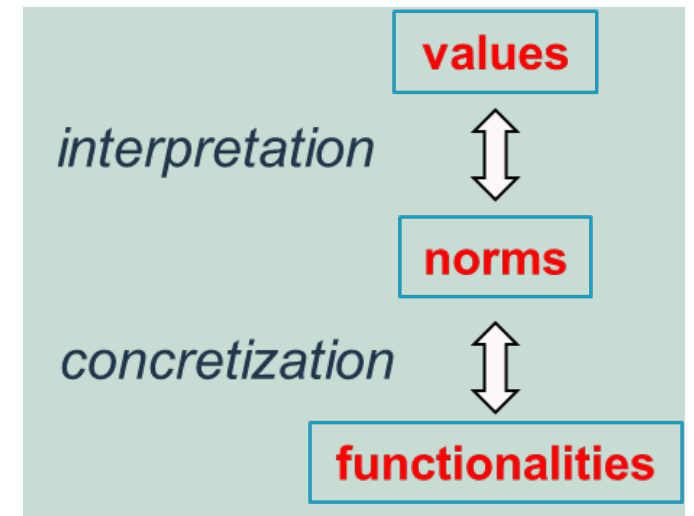
- Is AI all that different from other technologies?
 - Overlap with tech ethics, business ethics, medical ethics...
- Some specific challenges to AI brought by:
 - **Autonomy.** If executive control is delegated, who is responsible for actions? How can we maintain adherence to human values in autonomous decision-making?
 - **Deployment in open environments.** If we cannot predict the context which the system will encounter, how do we ensure ethical behaviour?
 - **Generalisation.** If a decision is taken by generalising from data, what happens to our right to be seen as an individual? How to avoid reproduction and amplification of biases?

RAI – MORE THAN ETHICS

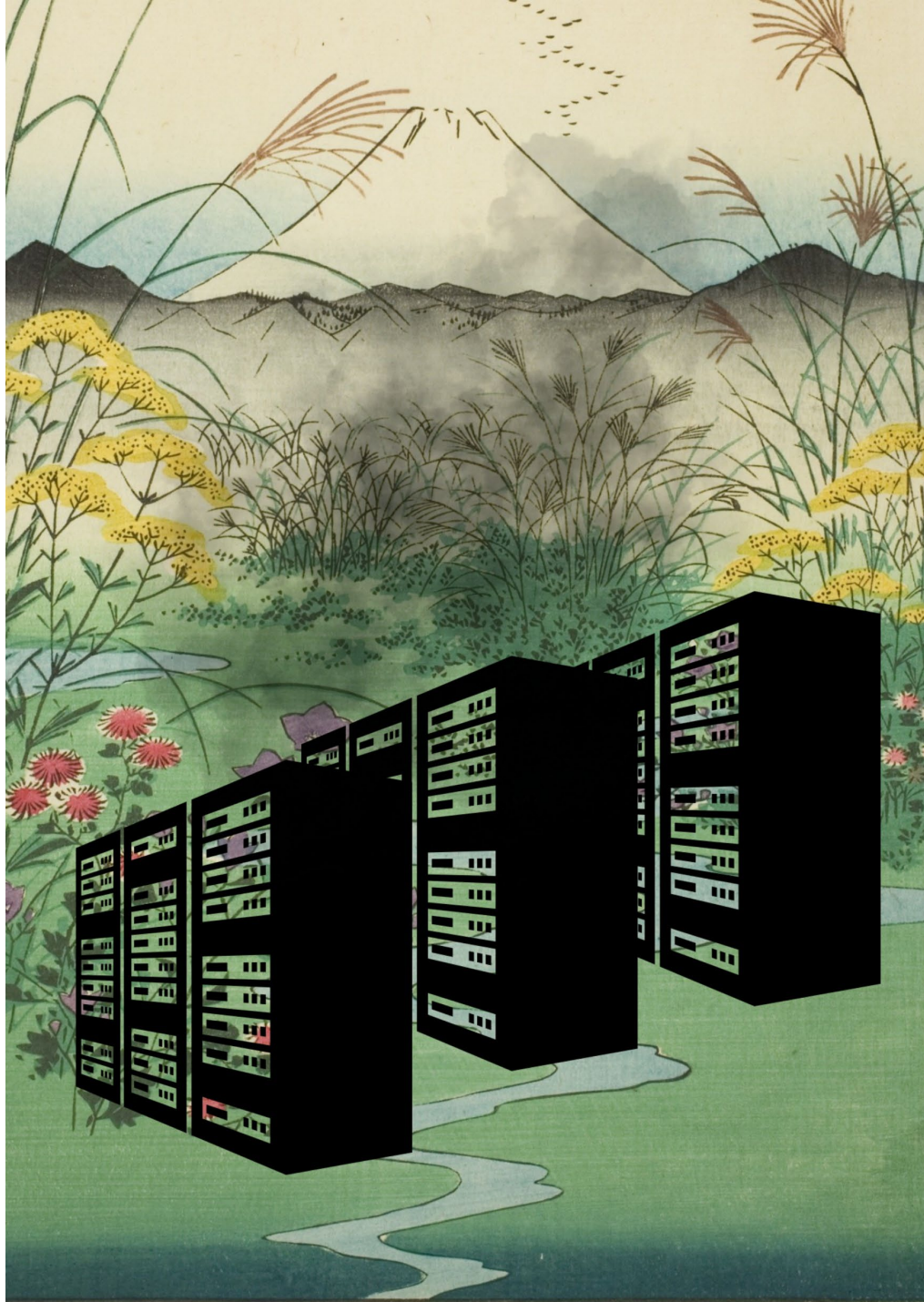
- Not philosophising about ethics
 - Ethics is not about the answer but about recognizing the issue
 - Ethics is a (social) process not a solution
- Not technification of ethics
 - Your implementation does not 'solve' ethics
- Fundamentally, is about choices, priorities, tradeoffs
 - Accuracy / Explanation
 - Accuracy / Computational resources
 - Security / privacy
 - Equity / equality
 - Long term benefit / Short term
 - ...

RAI IS ABOUT BEING EXPLICIT

- Design for Values
 - Legal and ethical aspects are not an add-on!
- Governance
 - External monitoring and control
 - Agreements, contracts, norms
- Design
 - Question your options and choices
 - Motivate your choices
 - Document your choices and options



<https://medium.com/@virginiadignum/on-bias-black-boxes-and-the-quest-for-transparency-in-artificial-intelligence-bcde64f59f5b>



A NEW AI PARADIGM

A NEW AI PARADIGM ?

- AI is not a universal solution
 - recognize its limitations in addressing complex challenges
 - Question zero: should AI be used here?
- Address inherent risks of bias and discrimination
 - improving data and algorithmic transparency
 - accommodate diverse contexts and values
- Beyond disciplines
 - integrate social and technical expertise in AI design
 - addressing systemic societal challenges
 - involving varied societal actors, not just technologists
- robust technical standards
 - Verifiable, sustainable, participatory

RESPONSIBLE AI APPROACH

Commit to responsible and trustworthy AI principles for innovative, ethical AI that respects human rights



Recognize potential tensions between ethical principles and the need to prioritize them



Develop a culture of responsibility within organizations



Prioritise stakeholder interaction and communication

TRUSTWORTHY GOVERNANCE

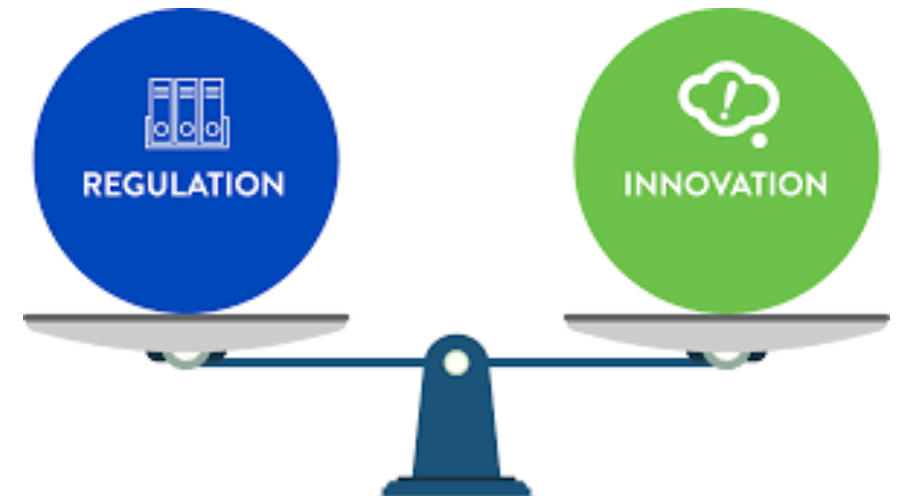
- Development of AI
 - What? Requirements, needs, possibilities, capabilities
- Use of AI
 - How? license to operate
- Context in which AI is developed and used
 - Where/When? rules of the game
- Global efforts
 - Sustainability
 - Inclusion and participation
 - Diversity
 - Distribution of benefits and costs
 - Agenda 2030 - SDGs



**political will and
innovation
needed!**

RESPONSIBLE AI IS INNOVATION

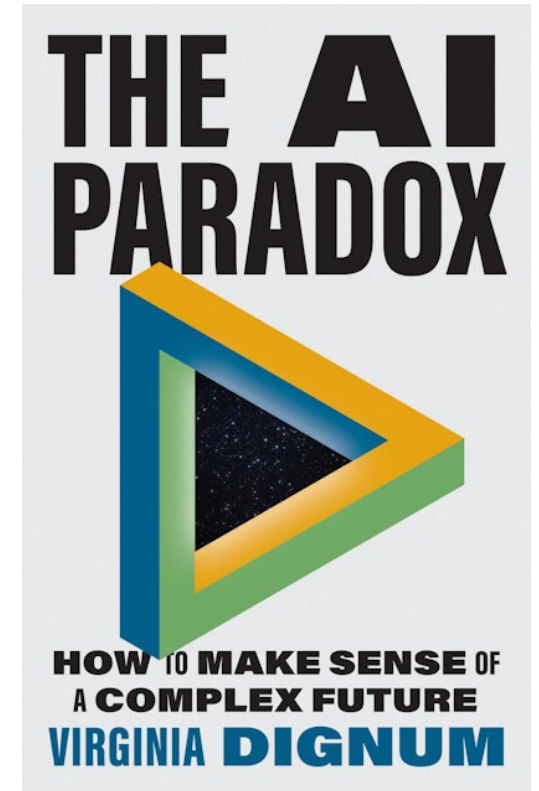
- Innovation needed
 - Technical requirements
 - Use requirements
 - Context of operation
- Multidisciplinary innovation!
 - Technological innovation
 - Organisational innovation
 - Regulation innovation
 - Governance innovation
 - Social innovation



AI is not an unstoppable force; it is shaped by our choices

- the decisions we make today will determine whether it becomes a tool for empowerment or a source of control and inequality.

AI's greatest paradox: the smarter it gets, the more we need human wisdom



Upcoming in February 2026

THANK YOU!